Annex I. Debt Sustainability Analysis

Lesotho remains at moderate risk of external debt distress. Under the baseline scenario, all indicators are well below their corresponding thresholds throughout the projection period, with the one exception of the present value of external debt-to-GDP ratio, which comes close to its threshold of 40 percent of GDP in 2025, mainly reflecting nonconcessional debt financing for a proposed large hydropower plant under the second phase of the Lesotho Highlands Water Project (LWHP-II). However, Lesotho's debt sustainability is sensitive to the scale and quality of this major project. Negative shocks to exports and the possibility of a large exchange rate depreciation also present risks to debt sustainability.

UNDERLYING ASSUMPTIONS

Lesotho's public debt sustainability analysis is based on the following assumptions:

Macroeconomic assumptions. The baseline scenario builds upon the medium-term macroeconomic projections agreed upon with the authorities during the 2015 Article IV consultation mission. Reflecting the current weakness in the manufacturing and construction sectors, along with the effects of the anticipated fiscal adjustment, real GDP growth is projected to drop to 2½-3 percent a year in 2015/16 and 2016/17. Growth is expected to rebound in following years, driven by the ramping up of production at a new diamond mine (Liqhobong) and the commencement of construction activities related to the water transfer component of LHWP-II. Over the long term, real GDP growth is projected to average about 3½ percent a year, which is relatively conservative compared with average growth of 4½ percent a year used in the previous DSA Update. The downward revision of the long-term growth rate reflects a more sluggish outlook for South Africa, Lesotho's key economic partner, and the assumption of a permanently lower level of SACU revenues, which would undercut public investment program.
Annual CPI inflation is expected to increase from 2.6 percent in 2014/15 to around 5 percent in the medium term. In addition, the baseline scenario assumes continued efforts to strengthen public sector efficiency, and sizeable fiscal adjustments in light of falling SACU revenue. Exports are expected to be mainly driven by diamonds while imports are subject to project related activities.

Fiscal assumptions. The fiscal year 2014/15 ended with a surplus of 0.6 percent of GDP. However the expected decline in SACU revenue poses a big challenge and would require greater commitment and efforts to maintain fiscal stability and sustainability going forward. The fiscal balance is expected to fall into a deficit of just over 3 percent of GDP in 2015/16 and 7½ percent of GDP in 2016/17 (assuming fiscal measures are taken). Beyond that an effective implementation of proposed reforms and fiscal measures should ensure a broadly balanced budget.

- Tax performance, especially for income taxes, is expected to increase only marginally as ongoing administrative measures continue to be implemented may result in better tax compliance.

- Non-tax revenue, mainly from mining sector is expected to increase due to increased production at Letseng and Liqhobong diamond mines and adoption of new Mining Policy.
- SACU revenue is expected to fall, to about 25½ percent of GDP in 2015/16 and 15½ percent of GDP in 2016/17, from about 29 percent in 2014/15. Given the prolonged slowdown in South Africa, much of the fall in SACU revenue is projected to be permanent, recovering only marginally in 2017/18 to little above 18 percent of GDP. It is expected to remain at that level over the medium term.
- Additional fiscal measures of 7 percent of GDP, cumulative, over the next three years are assumed. The bulk of the adjustments would come from expenditure side, requiring both expenditure compression and streamlining.
- Financing needs are expected to rise to about 2½ percent of GDP over the medium term and are expected to be met by a mix of domestic debt issuance and drawdown of government deposits.
- Government is expected to raise new domestic debt of about 4½ percent of GDP over the next three years in order to support a gradual fiscal adjustment.

Assumptions regarding Lesotho Highland Project Phase-II. The baseline assumptions regarding the hydropower component of LHWP-II remain broadly the same as those in the previous DSA Update, however, the timing has been push back to 2021–25. The project is assumed to cost about USD500 million with 80 percent financing from commercial sources. However, these assumptions are tentative as the project's feasibility study is still underway. Moreover, the assumptions on the scale, financing, and costs of the project have a significant impact on debt sustainability. A cost overrun and any deterioration in financing conditions could lead to external debt indicators breaching relevant threshold and endanger risk rating.

Interests Rate. A discount rate of 5 percent is used for present value. [1]

[1] See https://www.imf.org/external/np/pp/eng/2013/100413.pdf

Box 1. Analysis of Hydropower Component of LHWP-II[1]

Staff discussed with the authorities the macroeconomic effects of investing in the electricity sector, using a variant of the open-economy Debt, Investment, and Growth (DIG) model developed by Buffie, et al (2012). The base model—a full-fledged dynamic macroeconomic model that allows for investment in infrastructure and government financing through concessional, commercial, or domestic debt, and/or fiscal adjustment—was tailored to Lesotho. In particular, it incorporates investments in the electricity sector—investing in a pump storage hydroelectric power plant within the second phase of LHWP—which sells electricity to both foreign and domestic customers. In the model, increases in domestic energy supply are growth-enhancing while receipts from selling electricity abroad can ease the fiscal burden (tax increases), which is often associated with big investment project. Still, in the transition, debt as well as taxes can rise substantially to pay for this investment.

The model and the investment program were calibrated to capture key characteristics of Lesotho. The calibration matches Lesotho's data for initial conditions for the fiscal year 2020/21, when the project is set to start. It also captures the project's main challenges: (i) the costs for the project implementation; and (ii) whether South Africa will purchase enough electricity to make the plant run at a high capacity. Usually costs for similar projects are often underestimated—e.g., the Ingula pump storage scheme in South Africa is currently experiencing costs which are almost three times the feasibility study estimate. To reflect these costs differences, two cost scenarios were considered: a low-cost scenario where costs for the government budget amount to around one billion USD and a high -cost scenario which corresponds to actual costs being 2.5 times higher.[2]

The analysis revealed that the project can have substantial macroeconomic benefits in Lesotho, as long as low costs and a high electricity demand by South Africa are maintained. In the low-cost scenario, real GDP increases by 2.5 percent, while consumption increases by 5 percent in the long run. The gain in real GDP is supported by lower domestic electricity prices, which boost domestic production. Taxes increase by a moderate amount in the transition, which explains the small dip in consumption, during construction years (see Figure below). Public debt remains sustainable by increasing solely by 5 percentage points of GDP. In contrast, with higher costs, private consumption is significantly crowded out and stays below its pre-project level for almost two decades, since taxes have to increase by 5 percentage points and remain high throughout the simulation period. And although growth picks up more quickly in the short run, total public debt reaches almost 90 percent of GDP, raising concerns of debt distress and questioning the sustainability of growth prospects. Key in the analysis is the assumption of foreign demand by South Africa. If foreign customers do not agree to purchase enough electricity to keep production high, the results will deteriorate for both cost scenarios.

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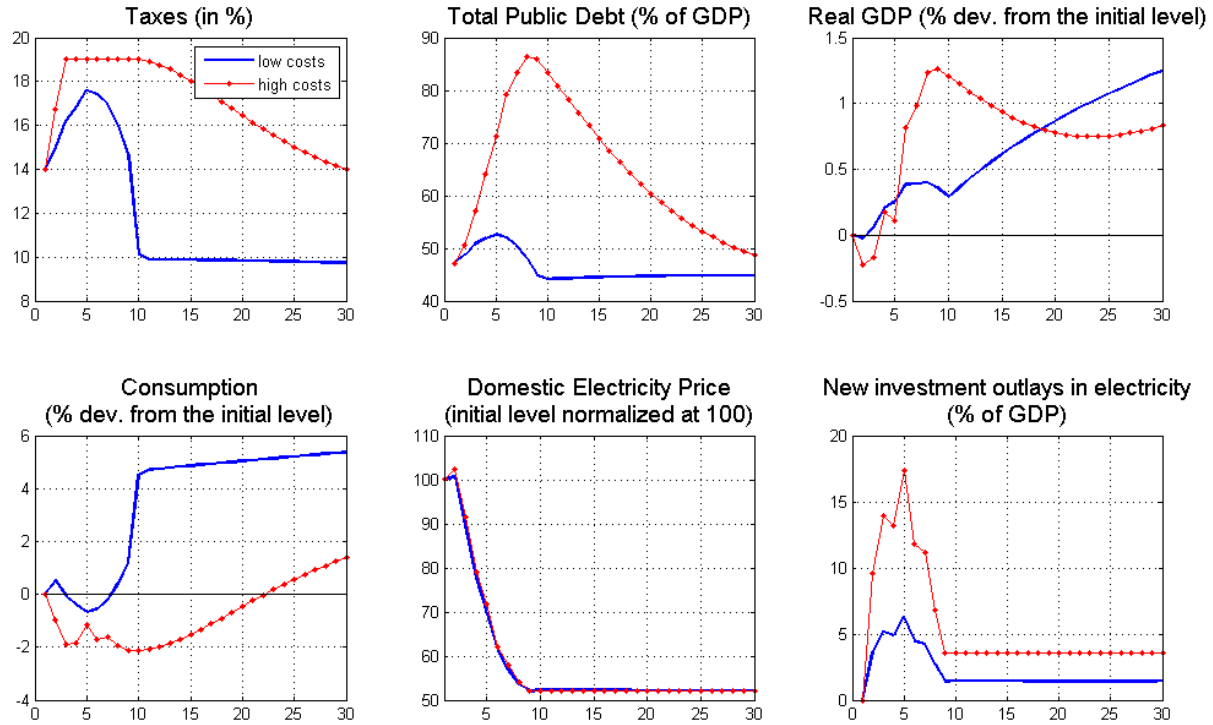
[1] Prepared by Michele Andreolli (RES).

[2] Pending the finalization of the feasibility study, DSA assumes the cost of the project at USD500 million (original project assumptions).

Box 1. Analysis of Hydropower Component of LHWP-II (concluded)



EXTERNAL DEBT SUSTAINABILITY

A. Baseline

All external debt sustainability indicators remain below their corresponding thresholds in the baseline scenario (Table 1a). For example, the present value (PV) of external debt to GDP ratio is expected to remain about 25 percent in medium term, after which time it is expected to rise to 37½ percent in 2025, before subsiding to 23½ percent by 2035. The temporary rise in the external debt trajectory is driven by financing for the construction of the hydropower plant under LHWP-II. Moreover, in the long run a prudent fiscal policy should ensure a stable debt outlook.

These results are sensitive to the assumptions mentioned before and in particular concern about the amount and concessionality of LHWP-II financing, the future of SACU revenues, and the overall fiscal management. The temporary increase in PV of external debt to GDP ratio is driven entirely by higher nonconcessional borrowing for the LHWP-II. The increased borrowing during the construction of the power plant does make Lesotho temporarily more vulnerable to external shocks. The authorities are currently undertaking a feasibility study, results of which would provide a better idea about the cost of the project and financing options. While the baseline assumes that the government revenue would decline from about 60 percent of GDP in 2014/15 to about 52 percent of GDP in 2018 and onwards, largely due to a decline in SACU revenue, a more

abrupt and permanent decline of these revenues—as discussed in the risk assessment matrix (Table 7) in the staff report for Lesotho's Article IV consultation—would require major fiscal adjustments over the next three years without risking macroeconomic stability.

Under the probability approach[2] (Figure 3), there exists a breach of the threshold for debt-to-GDP ratio in the baseline scenario. However, the breach is short-lived and not excessive. Again, this breach is mainly driven by the LHWP-II. This approach re-emphasizes the importance of ensuring that authorities should only undertake the project if it is financially and economically viable and implemented effectively with minimal cost overruns and extended delays. Taking this into consideration, staff remains of the view that Lesotho is still under the moderate risk of debt distress.

B. Sensitivity Analysis

Stress tests show that Lesotho's external debt vulnerabilities would increase significantly, in particular in the event of a major exchange rate depreciation, or if the export growth turned out to be lower than the historical average, (Table 1b). In addition, a 200 basis point increase in interest rates for new public sector loans in 2015–34 would increase the PV of external debt-to-GDP ratio to 47 percent in 2025. Similarly, in the event of a one-time 30 percent depreciation of the nominal exchange rate in 2015, the PV of external debt-to-GDP ratio would increase to 50 percent by 2024/25, gradually falling to below the threshold by 2030 and staying under it thereafter. In a scenario where export value grows at one standard deviation lower than the historical average, the PV of external debt-to-export ratio would rise to 47 percent by 2024. In a scenario in which the key variables are set at their average of the past 10 years, Lesotho's external debt ratios actually fall relative to the baseline, reflecting an average fiscal surplus over the past years thanks to the large SACU revenue, a decent GDP growth and inflation averaging about 5 percent. However, given the structural break, the historical scenario could be considered less relevant for the analysis.

PUBLIC SECTOR DEBT SUSTAINABILITY

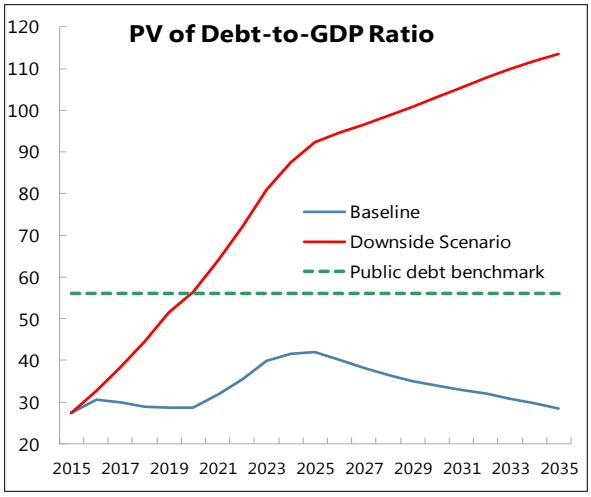
External debt accounts for more than 95 percent of Lesotho's public debt[3], public debt indicators largely mirror those of external debt. However a small increase in domestic debt (about 4 percent of GDP) is projected to facilitate gradual fiscal adjustments in coming years and expected to remain about 5 percent of GDP throughout the projected period.

[2] The probability approach methodology focuses on the evolution of the probability of debt distress over time, rather than on the evolution of debt burden indicators. This approach provides complementary, country specific information to help decide cases where a country's risk rating is on the border between two categories – in this case, a ±5 percent range is nearly breached from below for the present value of debt to GDP ratio in the baseline scenario and for the debt service to exports ratio under extreme shock.

[3] Public debt includes central government debt and debt guarantees, excluding potential contingent liabilities from state-owned entities.

The PV of public sector debt stood at below 30 percent in 2014/15 (Table 2a). The standard sensitivity tests for public debt distress point to higher degree of vulnerability of public debt to permanently higher deficit i.e. about 3 percent of GDP projected for 2015 (Table 2b). In addition, lower long-run GDP growth can also shift the debt trajectory higher but would still remain below the critical threshold of 60 percent of GDP.

To assess the implications of the risk of incomplete or delayed fiscal measures, an alternative scenario was undertaken (for details see Box 2 of the staff report). Given the higher fiscal primary deficits of about 6 percent of GDP per year and lower GDP growth, public debt vulnerability increases significantly in the medium term as the present value of debt-to-GDP ratio breaches the threshold of 56 percent in 2020 and continues to rise sharply afterwards. This again underscores the need for early and decisive actions on additional fiscal measures in order to avoid increasing the vulnerability of public debt.



CONCLUSION

Lesotho remains at moderate risk of external debt distress. All external debt and debt service indicators for the baseline remain below their respective thresholds. However, a large exchange rate depreciation or a decline in exports revenue pose major risk to Lesotho's debt sustainability. In addition, there is little scope for additional borrowing (beyond the baseline assumption) for financing the hydropower component of Lesotho High Lands Water Project-Phase II and findings of feasibility study would be important in this regard. The authorities are encouraged to seek bilateral and multilateral concessional sources of financing in order to minimize reliance on commercial borrowing.

The authorities broadly agreed with the assessment of moderate risk of external debt distress. They appreciated that the PV of external debt ratio rises owing to the construction of LHWP-II and agreed that care would be needed to ensure that the financing arrangements do not increase the risk of external debt distress. As part of their medium term debt strategy, the authorities intend to rely on concessional sources of external borrowing (with the exception of financing for LHWP-II). With the view to developing domestic debt market, they expect to expand issuance of government domestic securities.

Next DSA could usefully provide a more in-depth analysis of the extent of public debt vulnerabilities, including potential contingent liabilities, currency composition, and more updated data on LHWP-II. For example, in view of the recent developments with South Africa Rand, a continued exchange rate depreciation would increase debt service pressures for Lesotho.

Table 1a .Lesotho: External Debt Sustainability Framework, Baseline Scenario, 2012–35[1] (In percent of GDP, unless otherwise indicated)

	Actual			Historical 6/	Standard 6/	Projections									
	2012	2013	2014	Average	Deviation	2015	2016	2017	2018	2019	2020	2015-2020 Average	2025	2035	2021-2035 Average
External debt (nominal) 1/	**37.2**	**41.1**	**41.1**			**45.4**	**45.1**	**44.4**	**43.1**	**43.2**	**43.0**		**55.3**	**35.4**	
of which: public and publicly guaranteed (PPG)	37.2	41.1	41.1			45.4	45.1	44.4	43.1	43.2	43.0		55.3	35.4	
Change in external debt	37.2	3.9	-0.1			4.3	-0.3	-0.7	-1.4	0.1	-0.1		0.4	-2.2	
Identified net debt-creating flows	...	10.1	7.8			2.7	10.3	3.5	4.8	4.5	4.2		2.4	2.4	
Non-interest current account deficit	**9.9**	**10.2**	**7.7**	**-2.9**	**15.2**	**6.2**	**13.7**	**7.2**	**8.9**	**7.8**	**7.4**		**5.1**	**5.4**	5.7
Deficit in balance of goods and services	-155.8	-138.9	-138.9			-132.2	-130.4	-130.9	-130.1	-130.0	-125.6		-124.4	-122.6	
Exports	43.7	38.2	39.3			39.4	40.1	42.7	43.0	43.8	41.4		42.7	42.9	
Imports	-112.0	-100.7	-99.6			-92.8	-90.3	-88.2	-87.1	-86.2	-84.2		-81.7	-79.7	
Net current transfers (negative = inflow)	-40.0	-36.0	-37.2	-37.2	6.0	-33.0	-23.5	-26.8	-24.3	-24.3	-24.7		-24.2	-24.2	-24.2
of which: official	-33.1	-29.5	-30.9			-26.8	-17.4	-21.1	-20.0	-20.0	-20.4		-19.9	-20.0	
Other current account flows (negative = net inflow)	205.6	185.1	183.8			171.4	167.6	164.9	163.4	162.0	157.7		153.7	152.2	
Net FDI (negative = inflow)	**-2.4**	**-2.2**	**-2.1**	**-2.5**	**0.8**	**-2.6**	**-2.5**	**-2.2**	**-2.2**	**-2.2**	**-2.2**		**-2.2**	**-2.2**	-2.2
Endogenous debt dynamics 2/	**...**	**2.2**	**2.1**			**-1.0**	**-1.0**	**-1.5**	**-1.9**	**-1.1**	**-1.1**		**-0.6**	**-0.9**	
Contribution from nominal interest rate	...	0.5	0.6			0.1	0.3	0.4	0.4	0.4	0.4		1.2	0.3	
Contribution from real GDP growth	0.0	-1.4	-1.5			-1.1	-1.3	-1.9	-2.3	-1.5	-1.4		-1.8	-1.2	
Contribution from price and exchange rate changes	0.0	3.1	3.0			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**...**	**-6.2**	**-7.8**			**1.7**	**-10.6**	**-4.2**	**-6.2**	**-4.4**	**-4.3**		**-1.9**	**-4.5**	
of which: exceptional financing	...	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	21.5			24.8	25.0	24.9	24.3	24.5	24.6		37.6	23.4	
In percent of exports	...	...	54.7			62.9	62.4	58.3	56.5	55.9	59.4		87.9	54.6	
PV of PPG external debt	**...**	**...**	**21.5**			**24.8**	**25.0**	**24.9**	**24.3**	**24.5**	**24.6**		**37.6**	**23.4**	
In percent of exports	**...**	**...**	**54.7**			**62.9**	**62.4**	**58.3**	**56.5**	**55.9**	**59.4**		**87.9**	**54.6**	
In percent of government revenues	**...**	**...**	**36.8**			**43.9**	**51.9**	**48.5**	**47.6**	**47.9**	**48.0**		**73.4**	**45.8**	
Debt service-to-exports ratio (in percent)	**3.3**	**4.6**	**4.3**			**0.8**	**3.6**	**3.8**	**3.9**	**4.2**	**4.4**		**9.0**	**4.6**	
PPG debt service-to-exports ratio (in percent)	**3.3**	**4.6**	**4.3**			**0.8**	**3.6**	**3.8**	**3.9**	**4.2**	**4.4**		**9.0**	**4.6**	
PPG debt service-to-revenue ratio (in percent)	**2.5**	**3.2**	**2.9**			**0.6**	**3.0**	**3.2**	**3.3**	**3.6**	**3.6**		**7.5**	**3.9**	
Total gross financing need (Millions of U.S. dollars)	206.5	214.4	156.3			80.1	263.9	147.2	201.5	188.5	190.5		238.9	317.2	
Non-interest current account deficit that stabilizes debt ratio	-27.4	6.2	7.8			1.9	14.0	7.8	10.3	7.7	7.6		4.7	7.6	
Key macroeconomic assumptions															
Real GDP growth (in percent)	5.3	3.6	3.4	4.6	1.1	2.6	2.9	4.5	5.5	3.6	3.6	3.8	3.5	3.5	3.5
GDP deflator in US dollar terms (change in percent)	-11.4	-7.6	-6.9	0.9	10.4	-7.7	0.3	2.0	2.1	1.8	2.8	0.2	2.0	2.0	2.0
Effective interest rate (percent) 5/	...	1.3	1.3	1.3	0.0	0.3	0.7	1.0	0.9	0.9	0.9	0.8	2.4	0.9	1.5
Growth of exports of G&S (US dollar terms, in percent)	-10.5	-16.4	-0.8	2.3	13.0	-5.1	5.0	13.5	8.6	7.4	0.5	5.0	5.0	5.6	5.8
Growth of imports of G&S (US dollar terms, in percent)	-2.7	-14.0	-4.8	3.0	11.4	-11.7	0.5	4.1	6.4	4.3	4.0	1.3	5.1	5.6	5.2
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	44.7	42.3	42.3	42.2	42.3	42.1	42.7	26.2	42.3	34.6
Government revenues (excluding grants, in percent of GDP)	57.9	55.6	58.5			56.5	48.2	51.3	51.2	51.2	51.2		51.2	51.2	51.2
Aid flows (in Millions of US dollars) 7/	273.1	170.5	149.5			116.3	120.5	148.8	149.8	172.7	181.7		244.9	274.9	
of which: Grants	198.8	105.3	43.8			74.3	75.1	76.6	77.9	79.4	81.6		107.1	184.1	
of which: Concessional loans	74.4	65.3	105.7			42.0	45.4	72.2	71.8	93.3	100.0		137.8	90.9	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			4.6	4.5	4.8	4.5	4.7	4.6		4.6	3.7	4.2
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			80.1	78.2	72.0	72.1	68.8	67.8		51.5	80.9	64.6
Memorandum items:															
Nominal GDP (Millions of US dollars)	2308.4	2208.8	2127.3			2014.3	2079.4	2217.6	2389.1	2520.1	2681.8		3516.6	6046.9	
Nominal dollar GDP growth	-6.7	-4.3	-3.7			-5.3	3.2	6.6	7.7	5.5	6.4	4.0	5.6	5.6	5.6
PV of PPG external debt (in Millions of US dollars)			428.8			467.3	486.8	516.7	544.2	578.6	617.2		1236.7	1326.0	
(PVt-PVt-1)/GDPt-1 (in percent)						1.8	1.0	1.4	1.2	1.4	1.5	1.4	2.2	0.0	1.5
Gross workers' remittances (Millions of US dollars)	251.9	204.9	180.5			180.1	189.5	210.8	200.1	201.8	241.2		290.9	547.8	
PV of PPG external debt (in percent of GDP + remittances)	...	...	19.8			22.7	22.9	22.7	22.4	22.7	22.6		34.7	21.5	
PV of PPG external debt (in percent of exports + remittances)	...	...	45.0			51.2	50.8	47.7	47.3	47.3	48.8		73.6	45.1	
Debt service of PPG external debt (in percent of exports + remittance	...	...	3.6			0.7	2.9	3.1	3.3	3.6	3.7		7.5	3.8	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 1b. Lesotho: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2015–35

(In percent)

	Projections							
	2015	2016	2017	2018	2019	2020	**2025**	2035
PV of debt-to GDP ratio								
Baseline	25	25	25	24	24	24	**37**	23
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	25	12	5	-4	-11	-18	**-33**	-64
A2. New public sector loans on less favorable terms in 2015-2035 2	25	24	25	25	26	26	**47**	34
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	25	23	23	23	23	23	**35**	21
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	25	28	37	36	36	36	**46**	22
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	25	26	29	28	29	29	**43**	26
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	25	19	16	16	16	16	**29**	20
B5. Combination of B1-B4 using one-half standard deviation shocks	25	20	26	26	26	26	**39**	23
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	25	33	33	32	33	33	**49**	30
PV of debt-to-exports ratio								
Baseline	63	62	58	56	56	59	**87**	53
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	63	31	12	-9	-26	-44	**-77**	-149
A2. New public sector loans on less favorable terms in 2015-2035 2	63	60	57	57	58	64	**110**	80
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	63	58	54	53	52	55	**81**	49
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	63	81	130	125	123	129	**162**	77
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	63	58	54	53	52	55	**81**	49
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	63	46	37	36	36	38	**67**	48
B5. Combination of B1-B4 using one-half standard deviation shocks	63	53	72	70	69	73	**108**	64
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	63	58	54	53	52	55	**81**	49
PV of debt-to-revenue ratio								
Baseline	44	52	49	47	48	48	**73**	45
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	44	25	10	-7	-22	-36	**-65**	-125
A2. New public sector loans on less favorable terms in 2015-2035 2	44	50	48	48	50	51	**92**	67
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	44	48	46	45	45	45	**68**	41
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	44	57	72	70	70	70	**91**	43
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	44	54	57	55	56	56	**84**	51
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	44	39	31	30	31	31	**56**	40
B5. Combination of B1-B4 using one-half standard deviation shocks	44	42	51	50	51	51	**77**	46
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	44	69	65	63	64	64	**96**	58

Table 1b. Lesotho: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2015–35 (concluded)

(In percent)

	Projections							
	2015	2016	2017	2018	2019	2020	**2025**	2035
Debt service-to-exports ratio								
Baseline	1	4	4	4	5	5	**9**	5
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	1	4	4	3	3	3	**4**	-11
A2. New public sector loans on less favorable terms in 2015-2035 2	1	4	4	5	5	5	**7**	7
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	1	4	4	4	5	5	**9**	4
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	1	5	7	8	8	8	**16**	9
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	1	4	4	4	5	5	**9**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	1	4	4	4	4	4	**8**	4
B5. Combination of B1-B4 using one-half standard deviation shocks	1	4	5	6	6	6	**11**	6
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	1	4	4	4	5	5	**9**	4
Debt service-to-revenue ratio								
Baseline	1	3	4	4	4	4	**8**	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	1	3	3	3	3	2	**3**	-9
A2. New public sector loans on less favorable terms in 2015-2035 2	1	3	4	4	4	4	**6**	6
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	1	3	4	4	4	4	**8**	4
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	1	3	4	4	5	5	**9**	5
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	1	4	4	5	5	5	**10**	5
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	1	3	3	3	4	4	**7**	3
B5. Combination of B1-B4 using one-half standard deviation shocks	1	3	4	4	4	4	**8**	5
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	1	5	5	5	6	6	**11**	5
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	31	31	31	31	31	31	**31**	31

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 2a. Lesotho: Public Sector Debt Sustainability Framework, Baseline Scenario, 2012–35

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate					Projections				
	2012	2013	2014	Average 5/	Standard Deviation 5/	2015	2016	2017	2018	2019	2020	2015-20 Average	2025	2035	2021-35 Average
Public sector debt 1/	40.3	43.4	42.7			48.1	50.6	49.5	47.6	47.3	47.3		60.0	41.2	
of which: foreign-currency denominated	37.2	41.1	41.1			45.4	45.1	44.4	43.1	43.2	43.0		55.3	35.4	
Change in public sector debt	34.3	3.0	-0.7			5.4	2.6	-1.2	-1.9	-0.3	0.0		0.5	-2.1	
Identified debt-creating flows	...	4.7	0.3			5.4	5.8	-3.3	-3.7	-2.6	-2.4		-2.7	-1.9	
Primary deficit	-5.9	1.8	-1.3	-2.8	7.5	2.9	6.9	-0.8	-0.8	-0.7	-0.2	1.2	-1.0	-0.3	-0.6
Revenue and grants	66.5	60.4	60.6			60.2	51.8	54.8	54.4	54.4	54.2		54.2	54.2	
of which: grants	8.6	4.8	2.1			3.7	3.6	3.5	3.3	3.2	3.0		3.0	3.0	
Primary (noninterest) expenditure	60.5	62.2	59.2			63.1	58.7	53.9	53.6	53.6	54.0		53.2	53.9	
Automatic debt dynamics	...	2.9	1.6			2.5	-1.0	-2.4	-2.9	-1.8	-2.2		-1.6	-1.6	
Contribution from interest rate/growth differential	...	-1.4	-1.4			-1.2	-1.5	-2.4	-2.9	-2.0	-2.0		-1.6	-1.6	
of which: contribution from average real interest rate	...	0.0	0.0			-0.1	-0.2	-0.2	-0.4	-0.4	-0.3		0.4	-0.2	
of which: contribution from real GDP growth	-0.3	-1.4	-1.4			-1.1	-1.4	-2.2	-2.6	-1.7	-1.6		-2.0	-1.5	
Contribution from real exchange rate depreciation	...	4.3	3.0			3.7	0.5	-0.1	0.0	0.2	-0.2		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	...	-1.6	-0.9			0.0	-3.3	2.1	1.8	2.3	2.4		3.2	-0.1	
Other Sustainability Indicators															
PV of public sector debt	...	...	23.1			27.4	30.5	29.9	28.8	28.7	28.8		42.3	29.2	
of which: foreign-currency denominated	...	...	21.5			24.8	25.0	24.9	24.3	24.5	24.6		37.6	23.4	
of which: external	...	...	21.5			24.8	25.0	24.9	24.3	24.5	24.6		37.6	23.4	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	-3.7	4.0	0.7			3.4	9.0	1.7	1.8	2.0	2.5		3.9	3.0	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	38.2			45.6	58.9	54.6	53.0	52.7	53.2		78.0	53.8	
PV of public sector debt-to-revenue ratio (in percent)	...	...	39.5			48.5	63.3	58.3	56.4	56.0	56.3		82.6	57.0	
of which: external 3/	...	...	36.8			43.9	51.9	48.5	47.6	47.9	48.0		73.4	45.8	
Debt service-to-revenue and grants ratio (in percent) 4/	3.4	3.7	3.4			0.8	4.1	4.7	4.7	5.0	4.9		9.2	6.1	
Debt service-to-revenue ratio (in percent) 4/	3.9	4.1	3.5			0.9	4.4	5.0	5.0	5.3	5.2		9.7	6.5	
Primary deficit that stabilizes the debt-to-GDP ratio	-40.2	-1.2	-0.6			-2.5	4.3	0.3	1.1	-0.5	-0.2		-1.6	1.8	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	5.3	3.6	3.4	4.6	1.1	2.6	2.9	4.5	5.5	3.6	3.6	3.8	3.5	3.5	3.5
Average nominal interest rate on forex debt (in percent)	...	1.3	1.3	1.3	0.0	0.3	0.7	1.0	0.9	0.9	0.9	0.8	2.4	0.9	1.5
Average real interest rate on domestic debt (in percent)	7.3	1.9	3.3	4.7	2.6	4.3	4.9	3.8	3.8	4.2	4.2	4.2	4.2	4.2	4.2
Real exchange rate depreciation (in percent, + indicates depreciation	18.4	11.9	7.6	3.1	12.6	9.2	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	2.0	7.2	5.9	6.6	3.1	4.9	4.3	5.3	5.4	5.0	4.9	5.0	5.0	5.0	5.0
Growth of real primary spending (deflated by GDP deflator, in percer	2.9	6.4	-1.5	0.9	2.2	9.3	-4.2	-4.0	5.0	3.6	4.3	2.3	3.4	3.5	3.5
Grant element of new external borrowing (in percent)	...	...	...	...	...	44.7	42.3	42.3	42.2	42.3	42.1	42.7	26.2	42.3	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

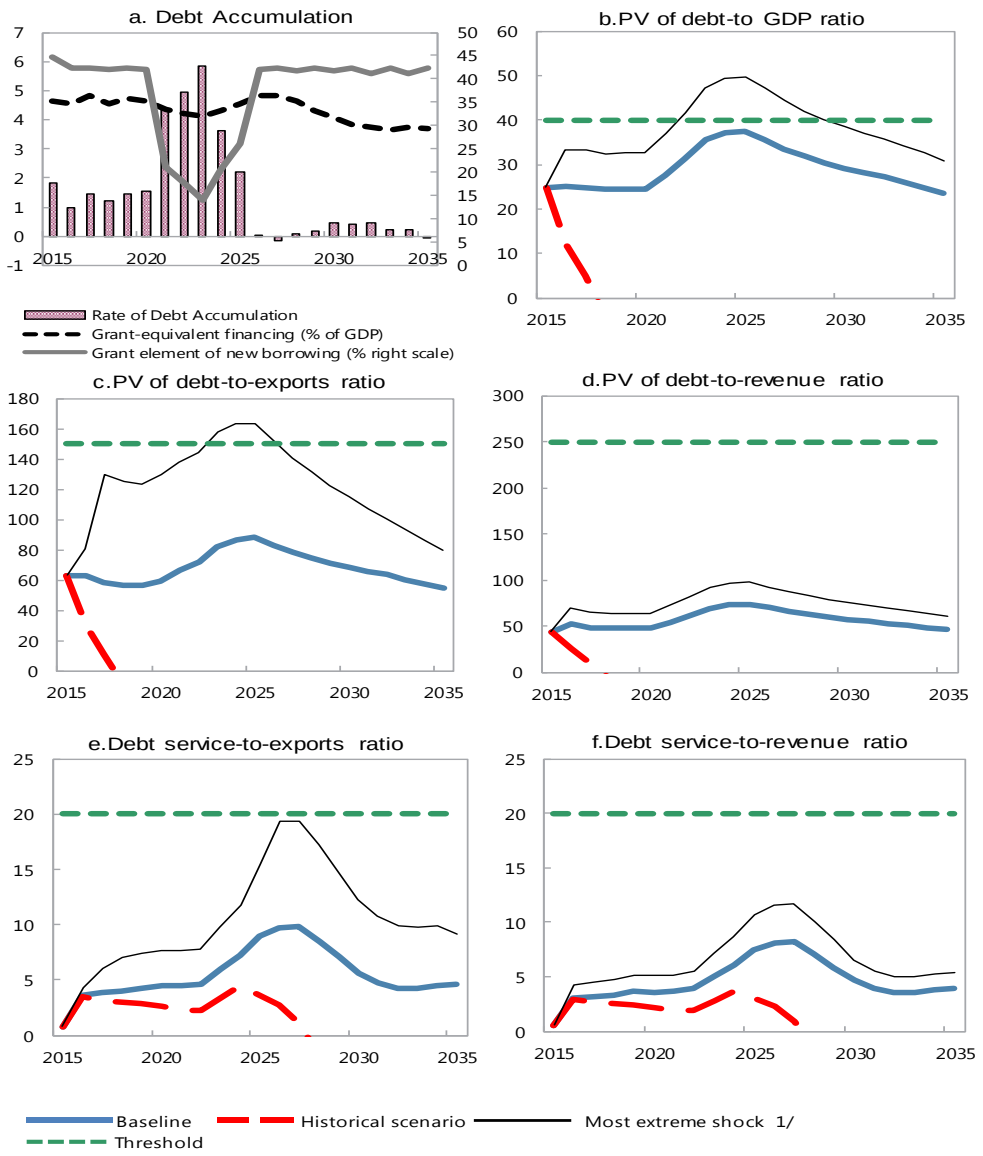
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2b. Lesotho: Sensitivity Analysis for Key Indicators of Public Debt 2015–35
(In percent)

	Projections							
	2015	2016	2017	2018	2019	2020	2025	2035
PV of Debt-to-GDP Ratio								
Baseline	27	31	30	29	29	29	42	29
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	27	24	22	20	19	17	23	-1
A2. Primary balance is unchanged from 2015	27	28	30	32	34	36	62	66
A3. Permanently lower GDP growth 1/	27	31	30	29	30	30	47	46
A4. Alternative Scenario : Insufficient Fiscal Measures	27	33	38	45	52	56	92	114
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2016-20	27	30	30	29	29	29	43	31
B2. Primary balance is at historical average minus one standard deviations in 2016-201	27	29	32	31	31	31	44	30
B3. Combination of B1-B2 using one half standard deviation shocks	27	26	27	26	26	26	38	24
B4. One-time 30 percent real depreciation in 2016	27	42	40	38	37	36	49	38
B5. 10 percent of GDP increase in other debt-creating flows in 2016	27	37	37	35	35	35	48	32
PV of Debt-to-Revenue Ratio 2/								
Baseline	46	59	55	53	53	53	78	53
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	46	46	40	37	34	31	42	-2
A2. Primary balance is unchanged from 2015	46	54	55	58	63	67	114	122
A3. Permanently lower GDP growth 1/	46	59	55	54	55	56	87	84
A4. Alternative Scenario : Insufficient Fiscal Measures	46	63	73	86	99	108	177	218
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2016-20	46	58	55	53	53	54	80	57
B2. Primary balance is at historical average minus one standard deviations in 2016-201	46	56	59	58	57	57	82	56
B3. Combination of B1-B2 using one half standard deviation shocks	46	51	50	48	47	47	70	44
B4. One-time 30 percent real depreciation in 2016	46	80	73	70	68	67	90	70
B5. 10 percent of GDP increase in other debt-creating flows in 2016	46	72	67	65	64	65	88	60
Debt Service-to-Revenue Ratio 2/								
Baseline	1	4	5	5	5	5	9	6
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	1	4	5	5	5	4	8	2
A2. Primary balance is unchanged from 2015	1	4	5	5	6	6	10	11
A3. Permanently lower GDP growth 1/	1	4	5	5	5	5	10	8
A4. Alternative Scenario : Insufficient Fiscal Measures	1	4	5	7	10	13	25	29
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2016-20	1	4	5	5	5	5	10	6
B2. Primary balance is at historical average minus one standard deviations in 2016-201	1	4	5	5	6	6	9	6
B3. Combination of B1-B2 using one half standard deviation shocks	1	4	5	5	5	5	9	5
B4. One-time 30 percent real depreciation in 2016	1	5	7	7	8	7	14	10
B5. 10 percent of GDP increase in other debt-creating flows in 2016	1	4	5	6	6	6	10	7

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

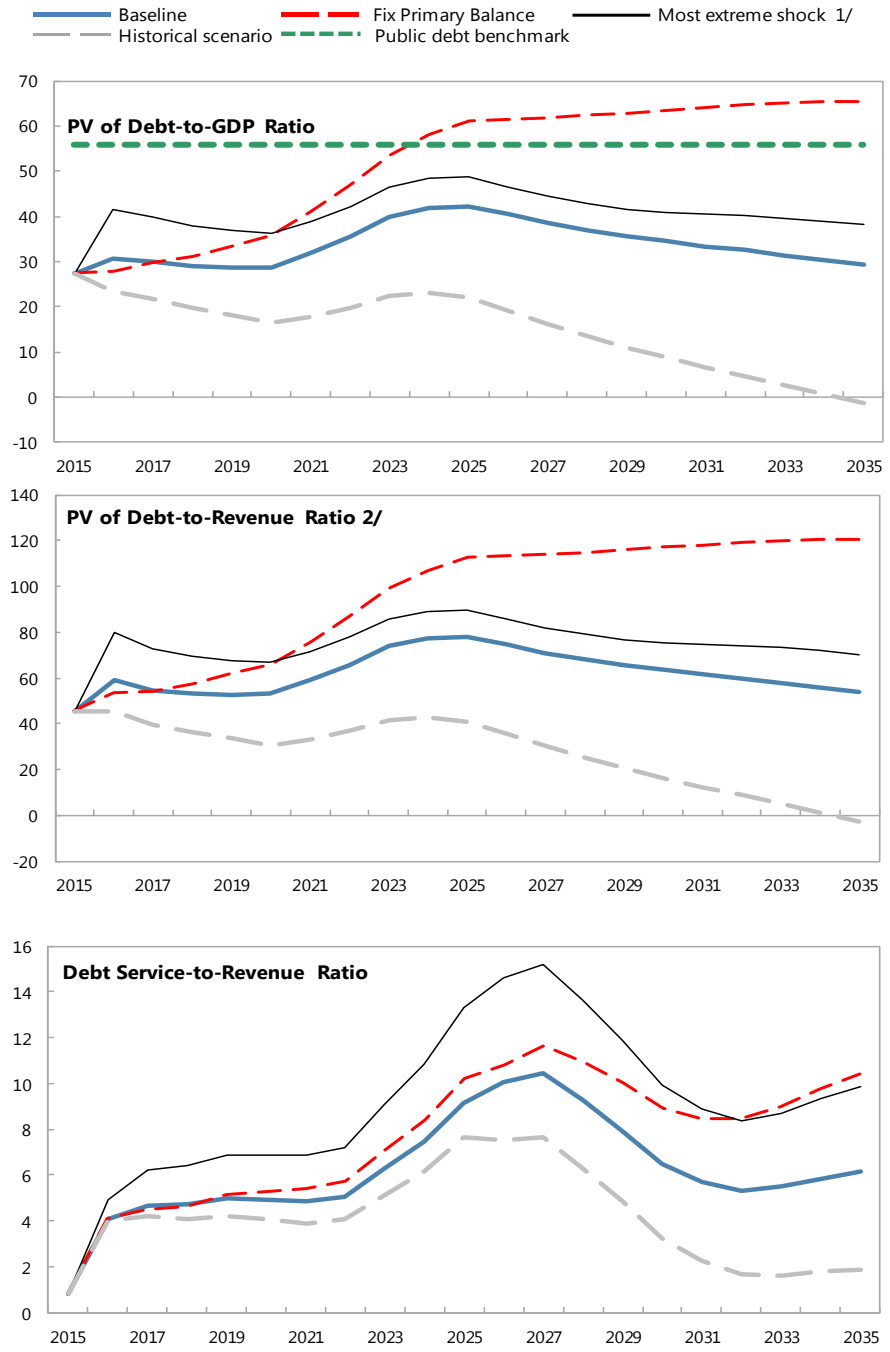
Figure 1. Lesotho: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2015–35[1]



Sources: Country authorities; and staff estimates and projections.

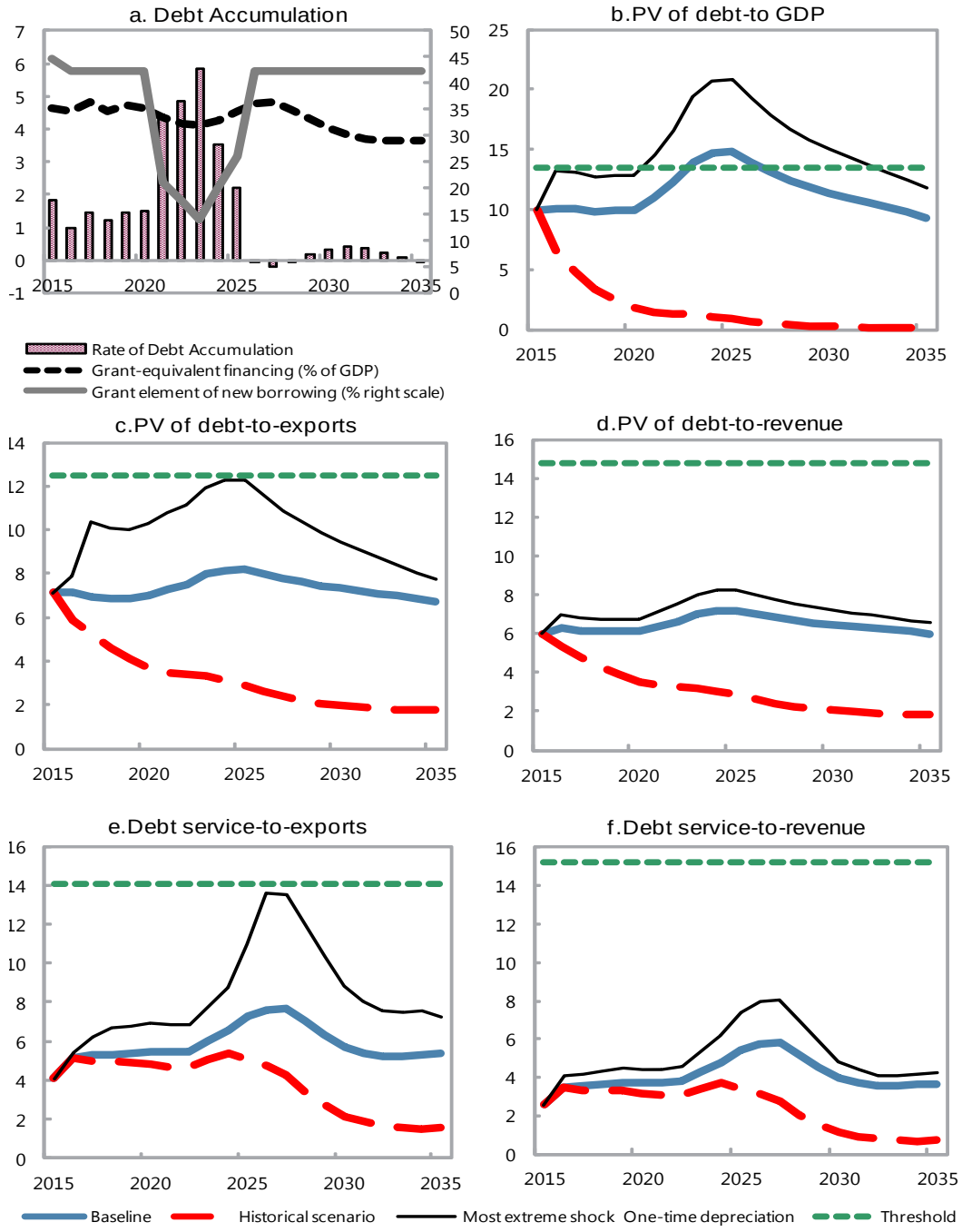
1/ The most extreme stress test is the test that yields the highest ratio on or before 2025. In figure b. it corresponds to a One-time depreciation shock; in c. to a Exports shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Figure 2. Lesotho: Indicators of Public Debt Under Alternative Scenarios, 2015–35[1]



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2025.
2/ Revenues are defined inclusive of grants.

Figure 3. Lesotho: Probability of Debt Distress of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2015–35[1]



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2025. In figure b. it corresponds to a One-time depreciation shock; in c. to a Exports shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock